Suraj Panda

Software Engineer | Backend & AI Systems

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WORK EXPERIENCE

Freelance Software Engineer *September 2026 – Present | Bengaluru, India*

- Designed and built a feature flag platform letting engineers roll out features gradually and enable or disable them at runtime, with no redeploy.
- Implemented a caching strategy for flag state in a Spring Boot backend, with a React admin dashboard for managing flags.

Technologies: *Spring Boot, MongoDB, Redis, React*

Software Engineer Intern – Full Stack | Aimerz *Apr 2025 – Jul 2025 | Bengaluru, India*

- Designed and implemented an atomic allocation engine for a multi-account savings platform, ensuring consistency under concurrent transactions using database-backed transactional guarantees.
- Wrote deterministic unit tests covering transaction rollback, JWT authorization flows, and concurrency edge cases, enabling safe refactoring of critical financial logic.

Technologies: *Next.js, Node.js, Express, PostgreSQL, Drizzle ORM*

PROJECTS

Inspect AI (UK AI Security Institute) | Merged PR GitHub

- **Fixed silent mis-clicks in Inspect AI's computer-use agent tool**, where out-of-range model coordinates were clamped to the screen edge. Added bounds checks to click, move, scroll, drag and zoom, validated against the screen size shown to the model instead of the native resolution.
- Returned invalid coordinates to the model as retryable tool errors, with **36** regression tests across **6** screen resolutions; merged after maintainer review.

Technologies: *Python, pytest, asyncio, Docker*

SafeSQL Proxy for LLM GitHub

- Prevented silent wrong answers from LLM-written SQL by building a semantic layer and MCP gateway that makes agents request metrics instead of SQL, ruling out fan-out and chasm inflation by construction.
- Hardened the request path with RBAC, row-level security, and CI checks that reconcile every metric against hand-written control queries and block any drop in natural-language query accuracy.

Technologies: *Go, PostgreSQL, MySQL, Snowflake, Databricks, DuckDB, MCP, OIDC, JWT, OpenTelemetry, Docker, Kubernetes, Helm, Elasticsearch, Tailwind*

Fault-tolerant Distributed Message Queue GitHub

- Built a Kafka-style message broker with partitioned topics, consumer groups with offset tracking, durable WAL storage, and configurable **at-least-once / at-most-once** delivery semantics.
- Implemented **Raft-based leader election and log replication** with quorum-based commits, follower catch-up, and crash recovery across multi-broker clusters.

Technologies: *Java, Spring Boot, Docker, Testcontainers, Prometheus, Grafana*

Semantic Cache for LLM GitHub

- Built a semantic caching library in Go with dual-path lookup ANN similarity search, reducing redundant LLM API calls by 30% across vector databases.
- Cut LLM request latency from **1.8s** to **250ms** on cache hits (86% improvement), benchmarked over 1,000+ operations.

Technologies: *Go, Prometheus, Grafana, Qdrant*

SKILLS

Languages: Go, Python, TypeScript, Java
Databases & Storage: PostgreSQL, MySQL, MongoDB, Redis, Qdrant
AI Coding Tools: Claude Code, OpenAI Codex
Tools & Technologies: React, Next.js, FastAPI, Node.js, Express, Spring Boot, MCP, Docker, PyTorch, Prometheus, Grafana, Playwright

EDUCATION

Siksha O'Anusandhan | Bhubaneswar, India *2019 – 2023*
Bachelor of Technology in Computer Science and Engineering | CGPA: 7.8/10

CERTIFICATIONS & ACHIEVEMENTS

- Ranked **#56 of 1983** in HackerRank hackathon, August 2026.
- LeetCode Knight (Rating: 2007); Global Rank 459 / 40,113 in Weekly Contest 510.
- AWS Certified Cloud Practitioner.